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Bucket Training

Build a bigram language model using physical tokens and buckets to track which words follow which other words.

You will need

- some text (e.g. a few pages from a kids book) printed or handwritten on paper (big enough that you can cut it into individual words)
- several containers (buckets are great, but could also cups, bowls, envelopes, or labelled areas on a table)
- scissors, pen and sticky notes or paper for bucket labels

Your goal

To produce a collection of labelled buckets containing tokens from your text.

Stretch goal: keep training your model on more input text.

Key idea

Language models learn by counting patterns in text. “Training” means building a model that tracks which words follow other words. In this version, the “following” relationship is captured physically — each bucket contains the tokens that appeared after its label in the original text.



Algorithm

1. prepare your tokens:

- print or write out your training text on paper
- use scissors to cut the text into individual words (called “tokens”); commas & full stops into separate pieces as well (and disregard all other punctuation)... but **keep them in order**

2. build the model one token at a time, starting with the first:

- if this token doesn’t have a bucket yet, create one and label it with this word
- take the *next* token from your pile and put it *into* the current token’s bucket
- now apply the same process to that next token (create its bucket if needed)
- repeat until all tokens are in buckets

Example

Original text: “See Spot run. See Spot jump.”

Prepared tokens: see spot run . see spot jump .

After processing the first two tokens (see spot):

Bucket	Tokens inside
see	spot

After all tokens have been processed:

Bucket	Tokens inside
see	spot spot
spot	run jump
run	.
.	see
jump	.

Notice that the “see” bucket contains two spot tokens because “spot” followed “see” twice in the original text. This captures the same information as a grid with tally marks, but in a physical form you can touch and manipulate.